

Chenwei Xu

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EDUCATION

Northwestern University, Ph.D. in Statistics

Evanston, IL, Sep. 2024–Jun. 2029

Northwestern University, M.S. in Computer Science

Sep. 2022–Mar. 2024

The Ohio State University, B.S. in Data Analytics

Aug. 2018–May 2022

TECHNICAL SKILLS

Methods	Multimodal/VLM evaluation, SFT/LoRA fine-tuning, reinforcement learning, 3D/4D world modeling, sparse Hopfield models
Systems	Benchmark construction, simulation-based evaluation, failure-mode analysis, ablation pipelines, model selection, reproducible reporting
Tools	Python, PyTorch, OpenMMLab, scikit-learn, Ray, Slurm, Docker, AWS, Linux, Git

RESEARCH EXPERIENCE

Dolby Laboratories

Aug. 2025–Present

Research Collaborator

Remote

- Drive applied research on VLM and video-generation evaluation, turning ambiguous failures into diagnostics for reasoning localization, temporal evidence selection, SFT/LoRA adaptation, and physical plausibility.
- Build benchmark and ablation workflows for VideoCritic, Phys4D, PhyPrompt, APT, and sparse video understanding, spanning simulation-grounded 4D evaluation, RL prompt refinement, and CVPR 2026 submissions.
- Produce paper-ready experiment reports, figures, and poster narratives that align claims, evidence, and next-step recommendations across remote multi-institution collaborators.

Northwestern University – Research Assistant

Evanston, IL, Sep. 2022–Present

- Led/co-led ML research across multimodal reasoning, tabular learning, sparse Hopfield memory, time-series forecasting, and scientific ML, resulting in first-author/co-first-author CVPR/ICML papers and broader NeurIPS/journal collaborations.
- Designed reproducible Python/PyTorch/Slurm/Docker experiment systems for OpenML tabular benchmarking, VLM diagnostics, RL-based accelerator control, antibody design, and high-dimensional model evaluation.
- Translated research ideas into benchmark definitions, model-selection pipelines, ablation studies, and publication-ready artifacts for BiSHop, Sparse Video Understanding, VideoCritic, AlignAb, Mu2e control, and related projects.

RESEARCH AREAS

- **Multimodal evaluation:** sparse video understanding, VLM reasoning-error localization, SFT/LoRA fine-tuning, physical plausibility, motion understanding, and 4D modeling.
- **Benchmark systems:** dataset/metric design, simulation-based evaluation, model diagnostics, failure-mode analysis, model-selection infrastructure, Slurm-scale execution, and ablation reporting.
- **Core ML methods:** BiSHop, sparse modern Hopfield models, tabular learning, feature programming, time-series forecasting, distributed HPO, and reinforcement learning.
- **AI for science:** antibody design, accelerator controls, beam-loss deblending, biomedical segmentation, and scientific-control workflows.

PUBLICATIONS

Video, LLMs & Multimodal Models

- **Towards Sparse Video Understanding and Reasoning.** [Chenwei Xu](#), Zhen Ye, Shang Wu, Weijian Li, Zihan Wang, Zhuofan Xia, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Han Liu. **CVPR**. 2026, Poster. [\[Link\]](#)
- **AdvEvo-MARL: Shaping Internalized Safety through Adversarial Co-Evolution in Multi-Agent Reinforcement Learning.** Zhenyu Pan, Yiting Zhang, Zhuo Liu, Yolo Y. Tang, Zeliang Zhang, Haozheng Luo, [Chenwei Xu](#), Yuwei Han, Jianshu Zhang, Dennis Wu, Hong-Yu Chen, Haoran Lu, Haoyang Fang, Manling Li, Chenliang Xu, Philip S. Yu, Han Liu. **ICML**. 2026, Poster.
- **VideoCritic: Diagnosing and Localizing Reasoning Errors in Video-Language Models.** [Chenwei Xu](#), Jianshu Zhang, Shang Wu, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Han Liu. **CVPR 2026 Workshop on Video Large Language Models (VidLLMs)**. Poster.
- **APT: Atomic Physical Transitions for Causal Video-Language Understanding.** Shang Wu, Haoran Lu, Songling Liu, [Chenwei Xu](#), Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Zhaoran Wang, Han Liu. **NeurIPS**. Under review.
- **MotionArena: Evaluating Fine-grained Motion Understanding for Manipulation in VLMs.** Yiming Liang, Jianshu Zhang, [Chenwei Xu](#), Weijie Yin, Han Liu. **Under review**. 2026.
- **Attention Sinks and Outliers in Attention Residuals.** Haozheng Luo, Haoran Dai, Shaoyang Zhang, Xi Chen, Eric Hanchen Jiang, Yijiang Li, Jingyuan Huang, Chenghao Qiu, [Chenwei Xu](#), Zhenyu Pan, Haotian Zhang, Binghui Wang, Yan Chen. **Under review**. 2026.
- **Phys4D: Fine-Grained Physics-Consistent 4D Modeling from Video Diffusion.** Haoran Lu, Shang Wu, Jianshu Zhang, Maojiang Su, Guo Ye, [Chenwei Xu](#), Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Zhaoran Wang, Han Liu. **CVPR 2026 Workshop on 4D**

Vision (4DV). Poster. [\[Link\]](#)

- **PhyPrompt: RL-based Prompt Refinement for Physically Plausible Text-to-Video Generation.** Shang Wu, [Chenwei Xu](#), Zhuofan Xia, Weijian Li, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Han Liu. *arXiv*. 2026. [\[Link\]](#)
- **Let's Try Again: Eliciting Multi-Turn Reasoning in Language Models via Simplistic Feedback.** Licheng Liu, Zihan Wang, Linjie Li, [Chenwei Xu](#), Yiping Lu, Han Liu, Avirup Sil, Manling Li. *ICML AI for Math Workshop*. 2025.
- **Open-Ended Multi-Modal Relational Reasoning for Video Question Answering.** Haozheng Luo, Ruiyang Qin, [Chenwei Xu](#), Guo Ye, Zening Luo. *IEEE RO-MAN*. 2023. [\[Link\]](#)

Tabular Learning, Hopfield Models & Time Series

- **SMUTF: Schema Matching Using Generative Tags and Hybrid Features.** Yu Zhang, Di Mei, Haozheng Luo, [Chenwei Xu](#), Richard Tzong-Han Tsai. *Information Systems*. 2025. [\[Link\]](#)
- **BiSHop: Bi-Directional Cellular Learning for Tabular Data with Generalized Sparse Modern Hopfield Model.** [Chenwei Xu*](#), Yu-Chao Huang*, Jerry Yao-Chieh Hu*, Weijian Li, Ammar Gilani, Hsi-Sheng Goan, Han Liu. *ICML*. 2024. [\[Link\]](#)
- **On Sparse Modern Hopfield Model.** Jerry Yao-Chieh Hu, Donglin Yang, Dennis Wu, [Chenwei Xu](#), Bo-Yu Chen, Han Liu. *NeurIPS*. 2023.
- **Feature Programming for Multivariate Time Series Prediction.** Alex Reneau, Jerry Yao-Chieh Hu, [Chenwei Xu](#), Weijian Li, Ammar Gilani, Han Liu. *ICML*. 2023. [\[Link\]](#)
- **SWGA: A Distributed Hyperparameter Search Method for Time Series Prediction Models.** Weijian Li, Haozheng Luo, [Chenwei Xu](#), Han Liu. *OpenReview*. 2024. [\[Link\]](#)

AI/ML for Science

- **AlignAb: Pareto-Optimal Energy Alignment for Designing Nature-Like Antibodies.** Yibo Wen, [Chenwei Xu](#), Jerry Yao-Chieh Hu, Kaize Ding, Han Liu. *NeurIPS*. 2025. [\[Link\]](#)
- **GenoArmory: A Unified Evaluation Framework for Adversarial Attacks on Genomic Foundation Models.** Haozheng Luo, Chenghao Qiu, Yimin Wang, Shang Wu, Jiahao Yu, Zhenyu Pan, Weian Mao, [Chenwei Xu](#), Haoyang Fang, Hao Xu, Han Liu, Binghui Wang, Yan Chen. *NeurIPS 2026 Evaluations and Datasets Track*. Under review.
- **Beyond PID Controllers: PPO with Neuralized PID Policy for Proton Beam Intensity Control in Mu2e.** [Chenwei Xu*](#), Jerry Yao-Chieh Hu*, et al.. *NeurIPS ML4Phy Workshop*. 2023. [\[Link\]](#)
- **Using Deep Learning for Automated Identification of Cone and Rod Photoreceptors.** Mengxi Zhou, Nathan Doble, Stacey S. Choi, Tianyu Jin, [Chenwei Xu](#), Srinivasan Parthasarathy, Rajiv Ramnath. *Biomedical Optics Express*. 2022. [\[Link\]](#)
- **FPGA Architectures for Distributed ML Systems for Real-time Beam Loss De-blending.** Michelle A. Ibrahim, et al. ([Chenwei Xu](#) included). *Fermilab Conference*. 2023. [\[Link\]](#)
- **Disentangling Beam Losses in the Fermilab Main Injector Enclosure using Real-Time Edge AI.** K. Hazelwood, J. Arnold, M. Austin, J. Berlioz, P. Hanlet, J.Y.C. Hu, M. Ibrahim, [Chenwei Xu](#), et al.. *JACoW*. 2023.
- **Edge AI for Accelerator Controls (READS): Beam Loss Deblending.** Jeremy Arnold, et al. ([Chenwei Xu](#) included). *FNAL Technical Report*. 2023.

Distributed Computing

- **Adaptive Batch Size Schedules for Distributed Training of Language Models with Data and Model Parallelism.** Tim Tsz-Kit Lau, Weijian Li, [Chenwei Xu](#), Han Liu, Mladen Kolar. *CPAL*. 2025. [\[Link\]](#)
- **Communication-Efficient Adaptive Batch Size Strategies for Distributed Local Gradient Methods.** Tim Tsz-Kit Lau, Weijian Li, [Chenwei Xu](#), Han Liu, Mladen Kolar. *arXiv*. 2024. [\[Link\]](#)

SELECTED PROJECTS

- **VideoCritic / Sparse Video:** build VLM error-localization and compact-evidence workflows for video-language reasoning.
- **Phys4D / PhyPrompt / APT:** develop SFT/LoRA, RL prompt-refinement, simulation-grounded 4D modeling, and physical-plausibility evaluations for video generation.
- **BiSHop / Sparse Hopfield:** build tabular learning pipelines with categorical/numerical embeddings, sparse memory layers, OpenML benchmarking, and training monitors.
- **Feature programming:** develop multivariate time-series forecasting methods and reproducible evaluation workflows.
- **AdvEvo-MARL / GenoArmory:** run safety, robustness, and adversarial-evaluation experiments for foundation and multi-agent models.
- **AlignAb:** build scientific ML workflows for Pareto-optimal energy alignment in antibody design.
- **Mu2e / beam-loss control:** implement PPO-based beam-intensity control and edge-AI workflows for accelerator monitoring.
- **Biomedical imaging:** build deep segmentation and evaluation workflows for cone and rod photoreceptor identification.

Service & Honors: Area Chair, ICLR 2026; Reviewer, NeurIPS / ICML / ICLR / CVPR / ECCV / AISTATS / COLM. **Awards:** ASA DataFest Finalist; SAE AI Mini-Challenge 2nd Prize; Veeam Software Endowed Scholarship.